BRAC University (Department of Computer Science and Engineering)
CSE 221 (Algorithms) for Spring 2026 Semester
(NO EXTRA PAGE IS ALLOWED)
Quiz 1
SET - A

Student ID: **Full Marks: 20** **Section:**
Name: **Duration: 25 minutes**

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1. Write down the time complexity of the following code snippet in **Big-O** notation. Here, n is an input variable and k is a constant. **(5 marks)**

```
for (int i=1; i*i<=n; i+=k)
{
   int sum = 0;
   for (int j=n; j>=1; j=j/k)
   {
      for (int m=1; m<=n; m+=3){
         if ( m%2==0) break;
         sum = sum + 10;
      }
   }
}
```

2. For attending the quiz, you have earned **2 marks**.

3. Write down the rules/steps/assumptions for converting a growth function to asymptotic time analysis. **(3 marks)**

4. You are given a **non-decreasingly sorted array** a of size n (elements may repeat), and two keys **key1** and **key2**. Find the largest segment **[index1, index2] using binary search in O(log n) time** such that

key1 <= a[index1] <= a[index2] <= key2

Sample Input:

Index	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**
Value	**4**	**5**	**5**	**8**	**8**	**8**	**9**	**10**	**11**	**23**	**23**	**25**

key1 = 8 key2 = 23
Answer: index1 = 4, index2 = 11
Explanation: The segment [4,11] contains elements that are >=8 and <=23.

Notes:
If several **key1** exist, then you need to report the **leftmost** index. Similarly, if several **key2** exist, you need to report the **rightmost** index.

Hints: Think of two independent binary search. **(10 marks)**

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Quiz 2

Student ID: **Full Marks: 20** **Section:**
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1. You are given an array A of n integers. The **absolute sum of a subarray is defined as the absolute value of the sum of its elements**. Using a Divide and Conquer approach, find the ***maximum absolute sum subarray*** among all possible non-empty contiguous subarrays of A. Your algorithm should run in O(n log n) time. **(10 marks)**

3	**4**	**-4**	**-3**	**-5**	**10**	**-3**	**0**	**-11**

In this example, the maximum sum subarray is [-4, -3, -5] where sum is |-4-3-5| = 12

3	**4**	**-4**	**-3**	**-5**	**10**	**-3**	**8**	**-11**

In this example, the maximum sum subarray is [10, -3, 8] where sum is |10-3+8| = 15

2. For attending the quiz, you have earned **2 marks.**
3. Show the simulation of the Partitioning step for the following array **(Pivot is 11)**. **(8 marks)**

Index	1	2	3	4	5	6	7	8
Value	6	10	13	5	8	9	11	21

Bonus (3 marks)

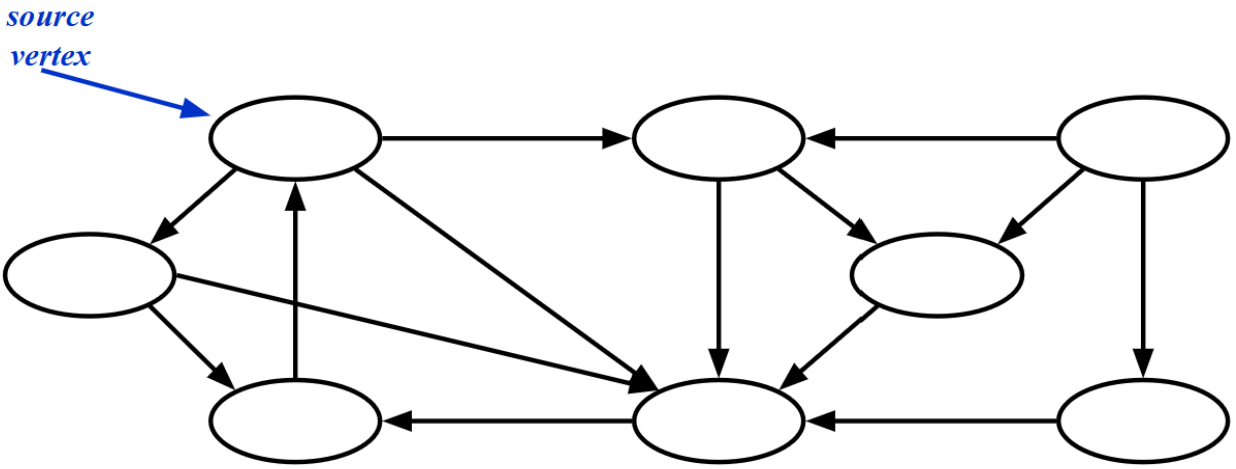
In **K-Way MergeSort**, an array of size n is divided into **k equal parts**, each part is recursively sorted, and then the k sorted subarrays are merged. What is the overall time complexity? Is it asymptotically better than standard MergeSort? Briefly justify.

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Quiz 4

Student ID: **Full Marks: 20** **Section: 20**
Name: **Duration: 15 minutes**

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1. Show the start and end times using DFS traversal. **(7 marks)**



2. For attending the quiz, you have earned **2 marks.**
3. Write down one usecase/application of the end times in DFS. **(2 marks)**

4. What do you mean by topological sorting? Give one example. **(4 marks)**

5. How to detect cycles in a graph? How will you modify the following code to determine if a cycle is present or not. **(3 marks)**

```
Bool cycle_present = false; // This variable should be set true if cycle is found
void dfs (int curr_node)
{
   color [curr_node] = 1;
   for ( int i = 0; i < adj_list[curr_node].size(); i++)
   {
       int next_node = adj_list[curr_node][i];
       [Fill in the blank here to detect cycle ……..……]
       if (color[next_node]==0) dfs(next_node);
   }
   color[curr_node]=2;
}
```

6. A graph always has a unique topological sorting. Do you agree or disagree? Explain your reasoning. **(2 marks)**

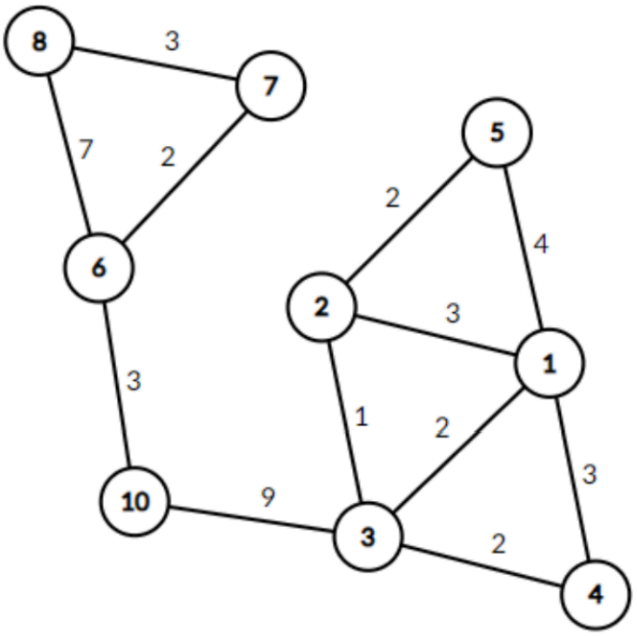
Bonus (2 marks)
If the start time of a node is n_s, end time of the node is n_e, and the start time and end time of any of its descendants are d_s and d_e respectively, what is the relationship among these 4 variables?

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Quiz 4

Student ID: **Full Marks: 20** **Section: 22**
Name: **Duration: 20 minutes**

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1. Construct the minimum spanning tree from the graph below. Present your work with proper details/data structures. Finally, draw the MST and write the total cost of that tree. **(10 marks)**



2. For attending the quiz, you have earned **2 marks.**

3. Welcome to the exciting realm of Bracula Gaming World, where you can earn lots of money by playing different games. But there's a twist – **time is limited to just 20 hours**! Each game has its own time limit and prize money if played for the full duration. However, **you can play for a fraction of the time and still earn money in proportion**. Check the list of games below. Your task is to make the most of your 20 hours and **collect as much money as possible**. **(8 marks)**

Game Name	Total Time Limit	Prize Money For Full Duration
Road Rush	15 hours	75$
GTA Vice City	20 hours	90$
Virtual Cop	3 hours	30$
House of the Dead	6 hours	55$
Far Cry	2 hours	17$
Devil May Cry	10 hours	47$

Write the names of the games and their respective playing times that will lead to **maximum prize money under the given constraints**. Show a simulation of your algorithm.